

Gastric cancer — Results & where to check

Every headline number below was independently re-computed against its source file (RESULTS_REVIEW.md). Verdict: all reproduce. Primary finding = an externally-validated stromal/fibroblast (CAF) programme; signature + microbiome + MR are secondary/exploratory and reported honestly (signature adds little over stage; microbiome tumour signal is batch-confounded; MR null).

Where to look

[PAPER.md](#)

Full manuscript — abstract, methods, results, discussion.

[RESULTS_REVIEW.md](#)

The source of this PDF: every result re-verified vs its file.

[results/figures/Fig1-4.pdf/.png](#)

The four main manuscript figures.

[FULL_REPORT_all_results_and_figures.pdf](#)

Pre-existing full report with all figures.

[VALID_RESULTS_cute.pdf](#)

One-page plain-English results sketchnote.

[results/](#)

31 result dirs / 165 tables — the raw source CSVs cited below.

[VERIFICATION_AUDIT.md](#) / [DEFECT_REPORT.md](#)

Rigor audit + corrections log.

Independent re-computation / re-reading of every `results/` subdirectory (31 directories, 165 tables) against the manuscript's claims. ****Verdict: every headline number reproduces,****
No unsupported claim, no fabrication signature, no duplicated-statistic artefact found.

Survival / prognostic signature

- **Nested-CV C-index**
0.611 (0.562–0.659); Uno 0.573
OK source: `nested\_cv/performance.csv`
- **Hartung-Knapp meta**
HR 1.19 (0.96–1.47), $p=0.073$, $I^2=19\%$
OK source: `meta\_HK/meta\_result.csv`
- **Per-cohort validation**
ACRG C=0.61 HR 1.90 (1.37–2.62); GSE15459 C=0.58 HR 1.68; GSE84437 null (C=0.53)
OK source: `validation\_multi/cindex\_HR\_summary.csv`
- **Time-varying HR**
1.49@12mo → 1.03@36 → 0.87@60; cox.zph $p=0.003$
OK source: `timevarying\_ACRG/\*`
- **External added value**
 $\Delta C +0.005$, IDI 0.021, NRI 0.175, LRT $p=0.002$
OK source: `external\_utility\_ACRG/added\_value\_external.csv`
- **Nomogram**
in-sample combined C=0.77–0.79; ACRG transported 0.66 / refit 0.72
OK source: `nomogram/\*`
- **Signature stability**
B=200 bootstrap; SERPINE1 retained; 13/25 >50%
OK source: `signature\_stability/\*`

Differential expression / enrichment

- **DEG integrated**
3,722 up / 4,025 down of 12,899; **** $\lambda=17.3$ ****
OK source: `tables/DEG\_integrated\_TCGA\_GTEEx.csv`
- **GTEEx-confound concordance**
top-100/200 = 100% replicated in TCGA-only
OK source: `tables/DEG\_TCGAonly\_replication.csv`
- **Hallmark EMT (diffuse)**
NES 3.17, padj 1.6×10^{-40}
OK source: `enrichment/GSEA\_Hallmark\_DiffuseVsIntestinal.csv`
- **Hallmark EMT (integrated tumour-vs-normal)**
positive NES
OK source: `enrichment\_integrated/fgsea\_Hallmark\_integrated.csv`

Immune / stroma / purity

- **Immune deconvolution**
T-cell(MCP) vs leukocyte% $\rho=0.67$, $p=3.6e-36$
OK source: `immune/validation\_vs\_measured.csv`
- **Tumour purity**
signature vs purity $\rho=-0.20$; vs stroma score +0.39
OK source: `purity/correlations.csv`

WGCNA / module preservation / single-cell

-
- **Red-module survival**
univariable HR 1.31 (1.12–1.53) $p=9.3e-4$; multivariable 1.35 (0.95–1.90) $p=0.090$
OK source: `wgcna\_real/ME\_survival\_cox\*.csv`
 - **Soft-power scale-free R^2**
corrected to 0.865–0.886 across powers 3–12
OK source: `wgcna\_real/soft\_threshold\_table.csv`
 - **Module preservation Z**
15.9 / 16.8 / 17.1
OK source: `module\_preservation/\*`
 - **scRNA composition**
43,992 cells; fibroblast 4.16%
OK source: `scrna/celltype\_composition.csv`
 - **scRNA localisation (non-circular)**
23/23 non-annotation hub genes fibroblast-dominant (median 0.96)
OK source: `scrna/gene\_dominant\_celltype\_noncircular.csv`

Microbiome (secondary, exploratory)

- **RF classifier**
cancer-vs-control AUC 0.916 (0.896–0.936)
OK source: `microbiome\_biomarker/05\_rf\_metrics\_and\_batch\_sanity.csv`
- **Batch leakage**
flowcell prediction 78% (baseline 55%)
OK source: same file
- **β -diversity collapse**
Bray R^2 0.065→0.011 after flowcell adjustment
OK source: `microbiome\_biomarker/02\_beta\_permanova.csv`
- **α -diversity effect sizes**
richness $\delta=-0.27$; Shannon -0.12 ; Simpson -0.07 (ns)
OK source: `microbiome\_biomarker/02\_alpha\_effectsizes\_cascade.csv`
- **Read-tracking fidelity**
Pearson $r=0.98$
OK source: `data/microbiome/reprocess/supp\_table3\_readtracking.csv`

Mendelian randomisation (secondary, exploratory)

- **MR IVW (all 6)**
all null; smallest $p=0.35$ (Streptococcus)
OK source: `mr\_real/MR\_results\_all\_methods\_REAL.csv`
- **Instruments**
all six at $p<1e-5$; F-range 19.3–20.3
OK source: `scratch/mr\_real.log`
- **MR-PRESSO global**
H. pylori borderline $p=0.052$; rest 0.16–0.68
OK source: `mr\_real/MR\_PRESSO\_global\_REAL.csv`
- **Egger intercept**
Fusobacterium 0.038, $p=0.040$ (nominal)
OK source: `mr\_real/MR\_pleiotropy\_REAL.csv`
- **East-Asian sensitivity**
all null (larger 7,921-case outcome)
OK source: `mr\_real\_eas/\*`

Drug repurposing (hypothesis-generating)

Top reverser hits are CDK4/6 (palbociclib) and PI3K/mTOR/FGFR (NVP-BEZ235, dovitinib, pd173074) inhibitors, targeting the proliferation program (FOXM1/TOP2A/MKI67/CDK1); consistent with §3.1–3.2 tumour-up biology. `drug\_repurposing\_integrated/candidate\_drugs\_ranked.csv`.

Orphan / unused outputs

`results/MaAsLin2\_gene\_vs\_microbiome/` — exploratory, no significant results, not referenced in the manuscript (now documented in PIPELINE.md).

****Conclusion.**** The manuscript is a faithful report of its result files. The only corrections this cycle were reporting/completeness items (documented in `DEFECT\_REPORT.md`), not data errors.

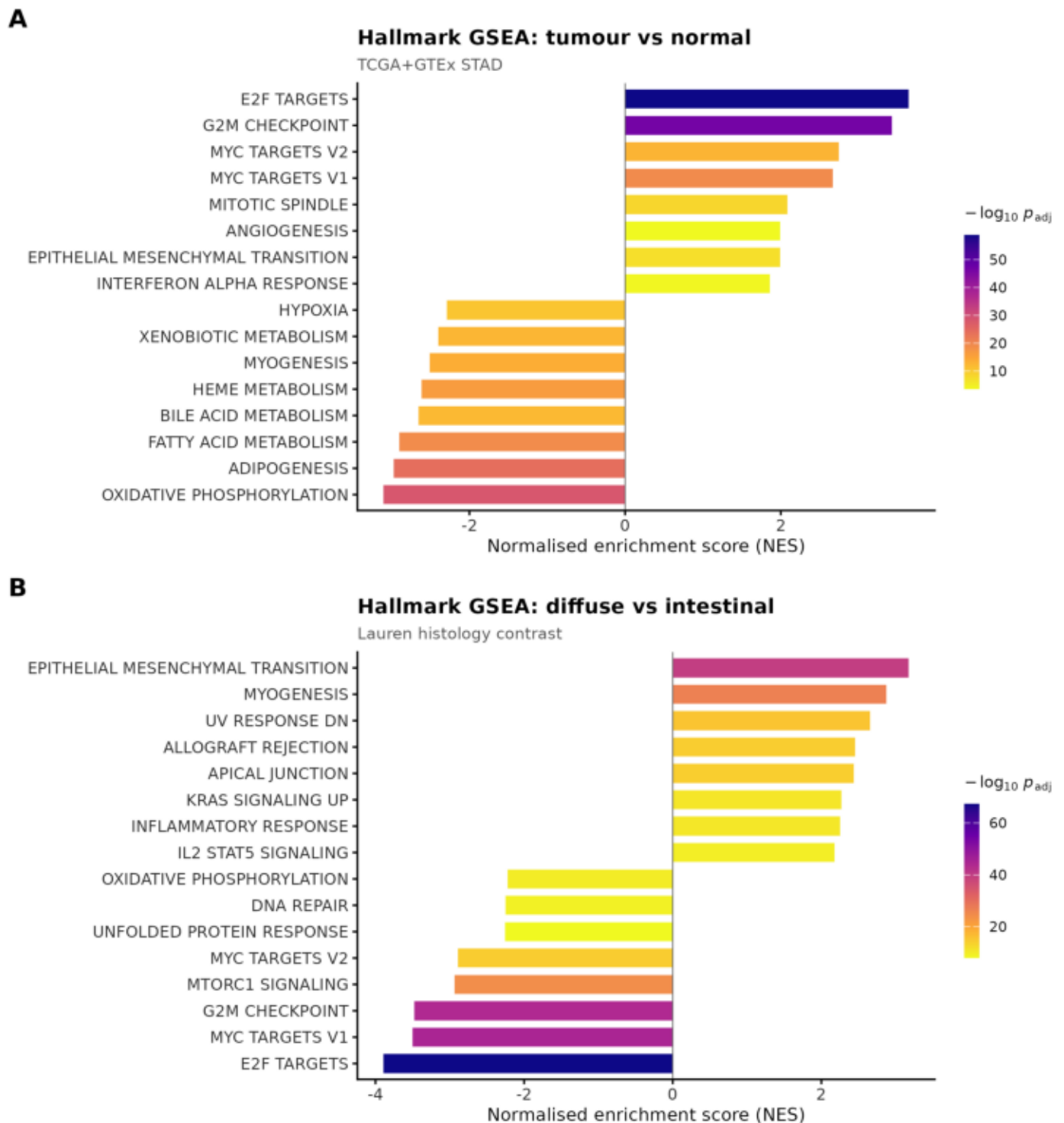


Figure 1 — Tumour transcriptional programmes. (A) Hallmark GSEA tumour vs normal; (B) diffuse vs intestinal (EMT/inflammatory up).

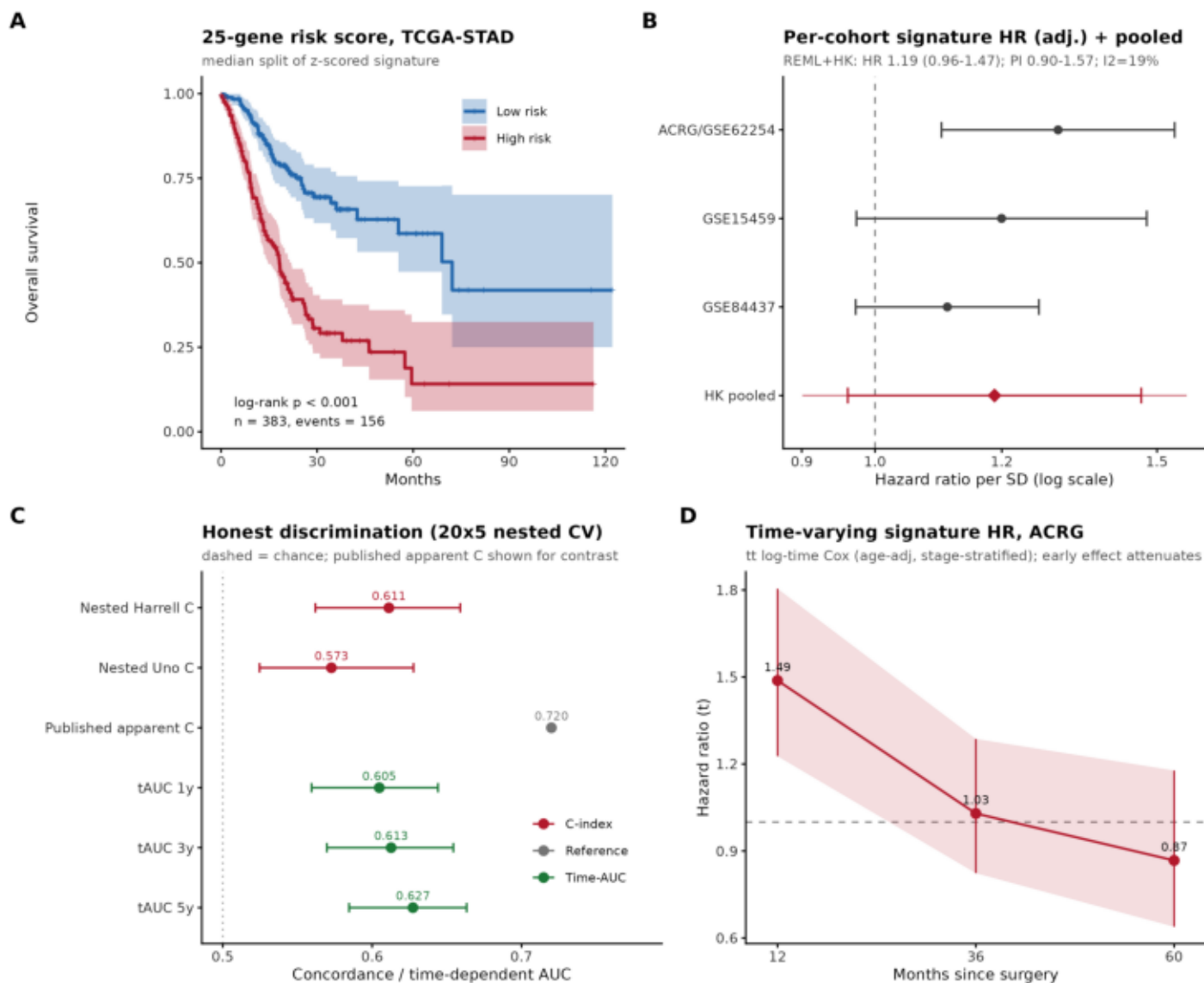


Figure 2 — The 25-gene signature, reported honestly. KM (TCGA), per-cohort forest + HK pooled HR 1.19 (ns), nested-CV C=0.611, ACRG time-varying HR.

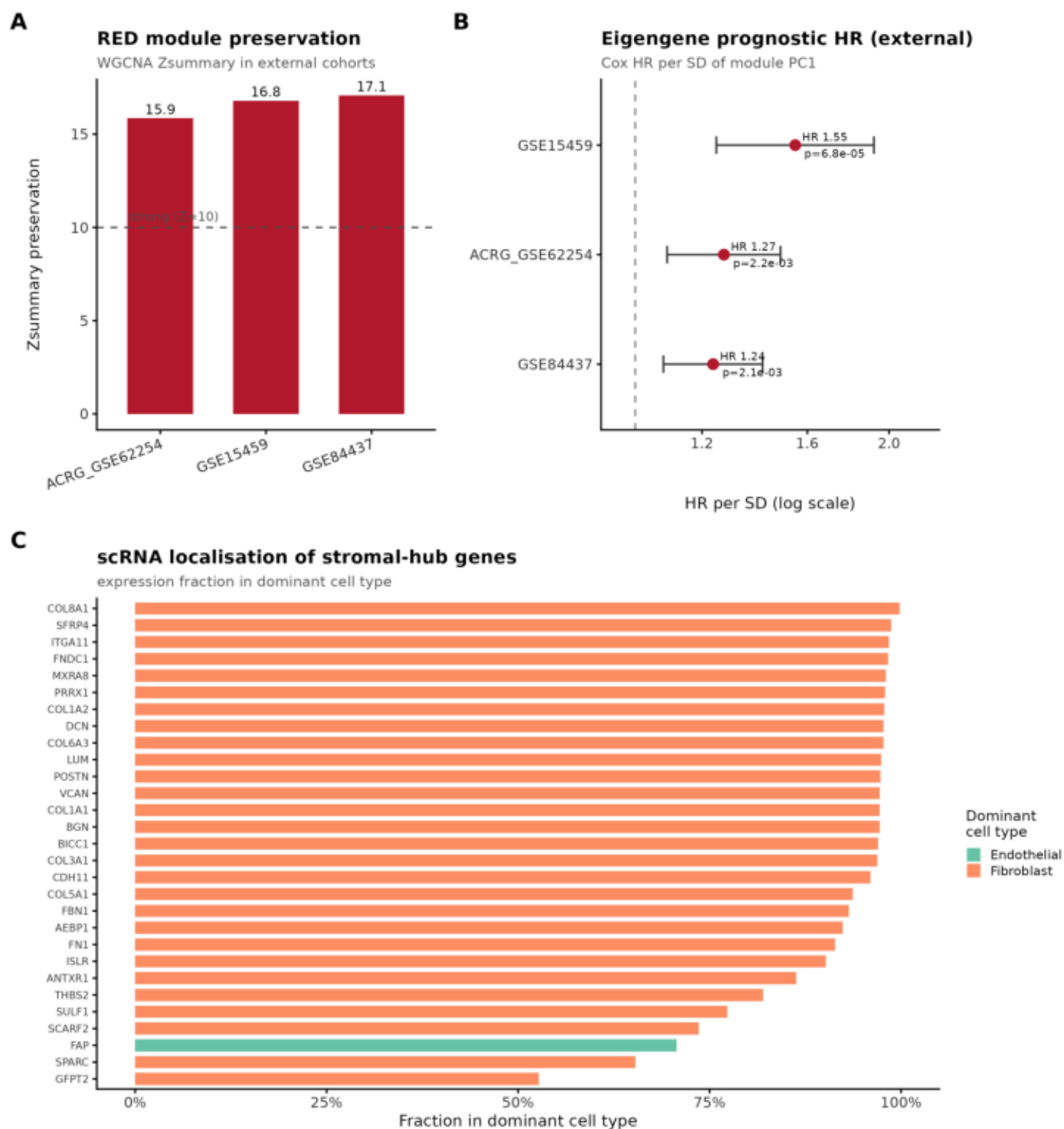


Figure 3 — Externally-validated fibroblast/stromal module (primary finding). Module preservation Z 15.9-17.1; external eigengene HR/SD 1.24-1.55; single-cell fibroblast localisation.

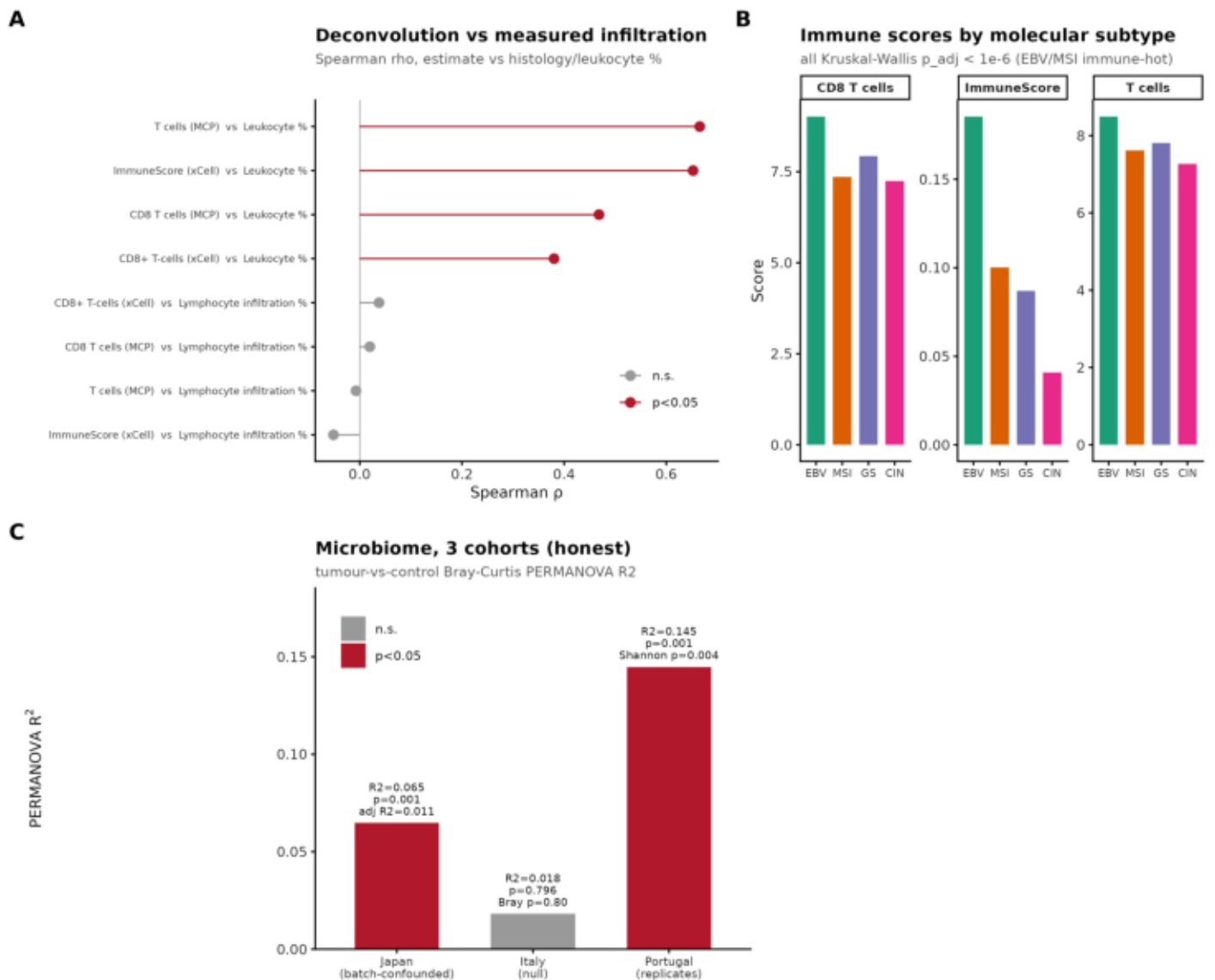


Figure 4 — Immune microenvironment + multi-cohort microbiome. Deconvolution validated vs measured leukocyte %; subtype immune scores; 3-cohort microbiome honesty panel (Japan batch-confounded).